

Enhancing Pipeline Efficiency : Payload & Deduplication

- **Focus:** Application Layer
- **Goal:** Lean & Unique Payloads
- **Impact:** Faster ingestion, lower compute costs

Lightening the Load: Local Refinement

Don't waste bandwidth on noise—compress, aggregate, and serialize efficiently.

BINARY OVER VERBOSE

Switch from heavy JSON to binary formats like **Protobuf** or **Avro**

SMART COMPRESSION

Use **Zstd** or **Snappy** to compress data locally before sending

SIGNAL OVER DATA

Send refined aggregates or state changes instead of raw telemetry

Refining the Cargo: Deduplication Strategy

Data duplication silently kills cloud performance. Ensure **idempotency** at the **source**.

BEYOND TIMESTAMPS

Assign a unique **ULID** or **UUID** to every event at the Edge

DE-DUPING AT SOURCE

Filter out redundant records before they ever hit the network

The *Data Engineering* Advantage: Making *UNION ALL* Fly

Clean, unique data at the warehouse enables:

ELIMINATE EXPENSIVE UNION

No more costly duplicate-removing operations

USE UNION ALL

Lightning-fast ingestion without “double counting”

SLASH COMPUTE COSTS

Reduce post-processing deduplication and associated CPU usage

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Data Engineer Insight:

A refined payload isn't just about saving bytes;
it's about **Data Integrity**. Lean & unique payloads at the Edge make every
optimization downstream more effective

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Data Engineer Insight: Reflect on Your Payload

Is your payload really refined, or are you just moving bytes? Does it ensure **Data Integrity**? Are your payloads at the Edge **lean and unique** enough to make downstream optimizations effective?

Vos payloads à la périphérie sont-ils assez **légers et uniques** pour rendre efficaces les optimisations en aval ?

Il tuo payload è davvero leggero e davvero ottimizzato per il *Cloud* ? Garantisce l'integrità dei dati ?